

REDDY UPPATHI

Senior Full Stack Engineer | AI-powered Development | GenAI

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Dubai, United Arab Emirates



SUMMARY

Senior Full Stack Engineer with 10+ years delivering high-scale apps, now specialising in AI-driven frontends – from on-device inference with TensorFlow.js to production RAG chatbots serving 1M+ users. Expertise includes **React, Angular, React Native, Node.js, MongoDB and GenAI** technologies.

WORK EXPERIENCE

Senior Full Stack Software Engineer 07/2023 - Present
Emirates NBD (ValueLabs) Dubai, UAE

- Developed an AI-powered conversational copilot widget using **React, Angular, React Native** and a **RAG pipeline**, handling **100K+ customer queries** per month and reducing live agent handoffs by 40%.
- Integrated a semantic search engine into the banking dashboard using **embeddings and vector similarity (Pinecone/pgvector)**, reducing support ticket resolution **time by 50%**.
- Led & Assisted the engineering of a shared UI component library leveraged across four projects, accelerating development cycles by 30%.
- Designed the agent's tool-use layer (5+ custom skills: transaction lookup, card block, fee reversal) and prompt instructions, achieving 99% intent routing accuracy across 200K+ MAU.
- Built an **intelligent form pre-fill module** leveraging on-device LLM inference (**Transformers.js**) to autopopulate fields from unstructured text, increasing user task completion rate by 25%.
- Built and scaled React applications for Emirates NBD, achieving 500K+ users and maintaining **99.5% crash-free** performance.
- Delivered an AI agent orchestration UI (React) that coordinated retrieval, fraud, and compliance agents over **MCP protocol**.
- Shipped a **real-time streaming RAG chat** (SSE + hybrid search) across web & mobile, achieved 91% answer relevance and 40% faster session times for **200K+ users**.
- Worked closely with product managers, UX designers, and analysts to deliver features that improved customer compliance by 30%.
- Streamlined CI/CD pipelines with Jenkins/GitHub, reducing deployment time by 20% for seamless releases.

Associate Consultant 01/2017 - 06/2023
ValueLabs Hyderabad, India

- Built unified cart/checkout APIs for retail POS + e-commerce, enabling 25% higher sales.
- Implemented code splitting, lazy loading, and memorization, reducing bundle size by ~35% and improving startup time.
- Reduced app crashes from 2.1% to 0.15% through comprehensive error boundary implementation and automated crash analytics.
- Led the team in building real-time fraud rules with Node.js/SQL, slashing **fraudulent transactions by 75%**.
- Contributed to reduce checkout latency by 60% by implementing Redis caching and PCI DSS-compliant tokenized payments.
- Developed & monitored an interactive, data-driven charts using D3.js to visualize sales data.
- Established comprehensive testing suites using Jest, React Native Testing Library, and Detox, achieving 85%+ unit and end-to-end test coverage.

Web Developer 07/2015 - 11/2016
Aptara Trivandrum, India

- Built a cancer e-learning platform (Node.js/Angular) for 10K+ doctors, **achieving 99.5% exam uptime**.
- Adapted and integrated WHO-compliant credit system certifying 5K+ oncologists yearly.
- Maintained 100% HIPAA and GDPR compliance, resulting in a **20% increase** in medical exam pass rates.
- Followed W3C Guidelines to ensure the application met Web Accessibility Compliance and Section 508 standards.

Junior Software Engineer 06/2014 - 06/2015
HurixDigital Hyderabad, India

- Developed a cloud-based e-learning platform, enabling educators to create 50% more interactive content.
- Debugged and Troubleshoot 20 UI/API P1 issues, improving platform stability and user satisfaction by 15%.
- Implemented responsive web design (RWD) with HTML5 and CSS3, ensuring cross-browser compatibility.

SKILLS

Frontend, AI & Machine Learning

React 17+ · Angular · Gen AI · TypeScript ·
JavaScript · React Native · ES6 ·
TensorFlow · Redux · D3 JS · LangChain ·
Embeddings · Vector Search ·
RAG Architecture · ONNX Runtime Web ·
Bootstrap · SCSS · CSS · Git · SonarQube ·
GraphQL · Flux · Next.js · Automation ·
Push Notifications · SDLC · Deep Linking

Backend & Databases

NodeJS · NestJS · Microservices ·
MongoDB · Swagger · Kafka · REST APIs ·
Pinecone · JWT · OAuth · SSE ·
Problem-Solving · Critical Thinking

Cloud & DevOps

AWS · S3 · EC2 · SQS · ELK · NGINX ·
Docker · CI/CD · Kubernetes · Agile · Jira

AGENTIC AI TOOLS

Claude Code · Cursor · Codex · Copilot

KEY ACHIEVEMENTS

- ★ AiDE Agents Certified Engineer
- ★ ValueLabs Best employee of the Quarter, Months (3 times)

COURSES

Machine Learning and Gen AI

Completed Machine Learning and Generative AI certifications from Udemy

EDUCATION

Bachelor of Technology

Jawaharlal Nehru Technological University
Hyderabad, India